اعمال ریاضی با اعداد

اعمال رياضي باينري: جمع

- قوانين: مانند جمع دسيمال
- با اين تفاوت كه 10 = 1+1 ← توليد نقلي
  - 0+0 = 0(c0) (sum 0 with carry 0)
  - 0+1 = 1+0 = 1(c0)
  - 1+1 = 0(c1)
  - 1+1+1 = 1(c1)

```
  +1 +1     +1
   1 0 1 1 0 1
+  0 1 1 0 0 1
-------------
 1 0 0 0 1 1 0
```

Carry	1	1	1	1	1	0
Augend	0	0	1	0	0	1
Addend	0	1	1	1	1	1
Result	1	0	1	0	0	0

سرریز (Overflow)

- اگر تعداد بیت ها = n و حاصل جمع n+1 بیت نیاز داشته باشد
  - ← سرریز

اعمال رياضي باينري: تفريق

- قوانين:
  - 0-0 = 1-1 = 0 (b0) (result 0 with borrow 0)
  - 1-0 = 1 (b0)
  - 0-1 = 1 (b1)
  - …

			0	10	1	1	10	10	
X	229		1	1	1	0	0	1	0	1
Y	– 46	–	0	0	1	0	1	1	1	0
$X - Y$	183		1	0	1	1	0	1	1	1

Borrow	1	1	0	0	
Minuend	1	1	0	1	1
Subtrahend	0	1	1	0	1
Result	0	1	1	1	0

روش انجام محاسبات

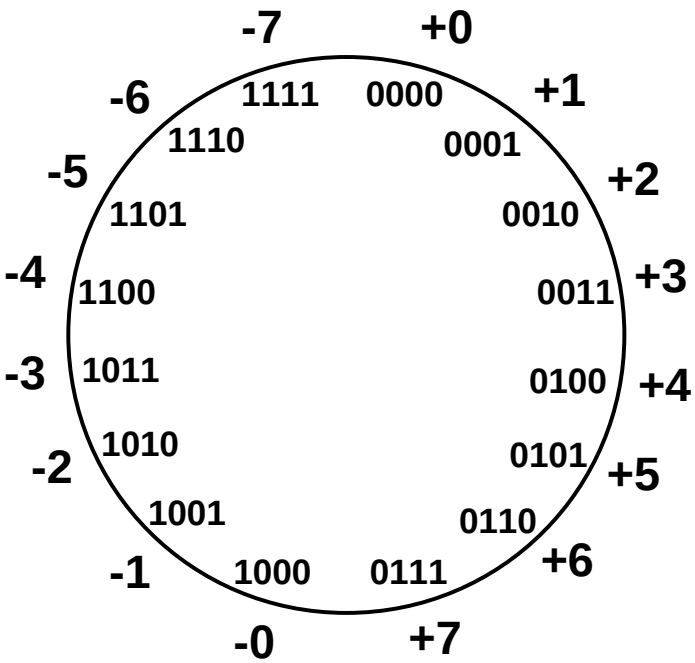
- الگوريتم هاي اعمال رياضي مبناي 10 را به خاطر آوريد.
- آنها را براي مبناي مورد نظر تعميم دهيد.
- قانون مبناي مورد نظر را به كار بريد.
  - براي باينري: 1+1=10

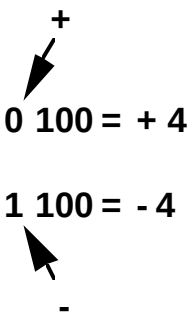
نمایش اعداد

- **نمایش اعداد مثبت:**
  - در بیشتر سیستم ها یکسان است.
- **نمایش اعداد منفي:**
  - اندازه-علامت (Sign magnitude)
  - مکمل 1 (Ones complement)
  - مکمل 2 (Twos complement)
    - در بیشتر سیستم ها: مکمل 2
- **فرض:**
  - ماشین با کلمه هاي 4 بیتي:
    - 16 ← مقدار مختلف قابل نمایش.
    - تقریباً نیمي مثبت، نیمي منفي.

نمایش اعداد

اندازه-علامت:





High order bit is sign: 0 = positive (or zero), 1 = negative

Three low order bits is the magnitude: 0 (000) thru 7 (111)

Number range for n bits = $[-(2^{n-1} -1), +(2^{n-1} -1)]$

Representations for 0

Cumbersome addition/subtraction

Must compare magnitudes to determine sign of result

نمایش اعداد

مکمل و:

N is positive number, then $\overline{N}$ is its negative 1's complement

$$\overline{N} = (2^n - 1) - N$$

Example: 1's complement of 7

$$\begin{array}{lcl} 2^4 & = & 10000 \\ -1 & = & \underline{00001} \\ & & 1111 \\ -7 & = & \underline{0111} \\ & & 1000 \end{array}$$

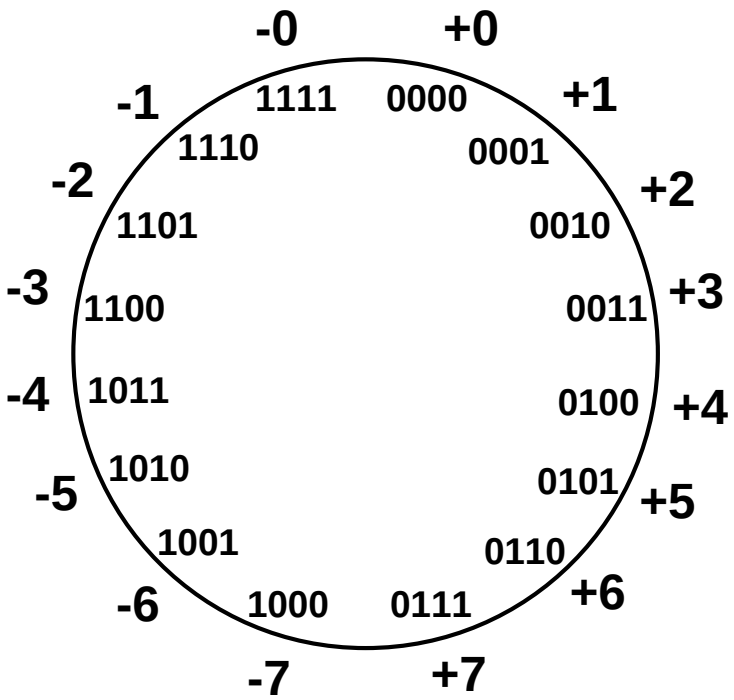
= -7 in 1's comp.

Shortcut method:

simply compute bitwise complement

0111 -> 1000

نمایش اعداد



È1 51m1

+

0 100 = + 4

1 011 = - 4

-

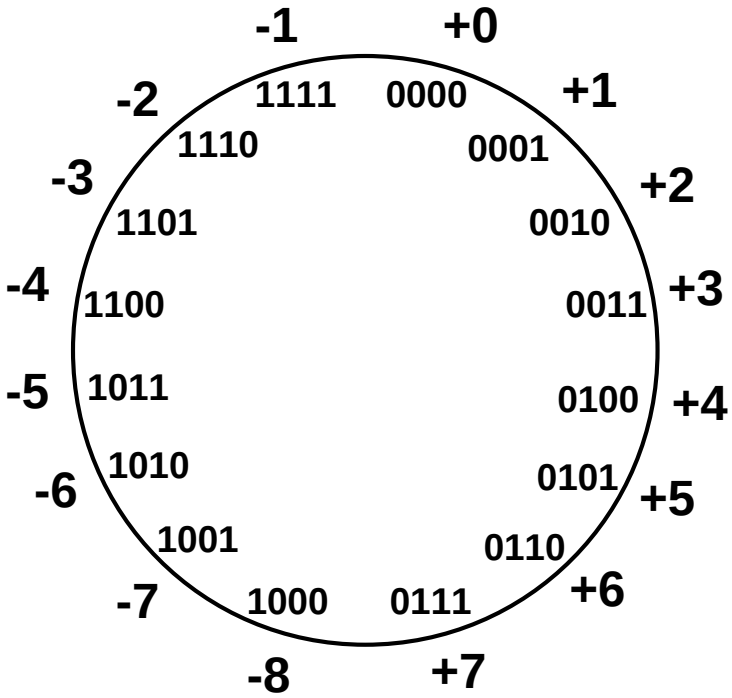
Subtraction implemented by addition & 1's complement

Still two representations of 0! This causes some problems

Some complexities in addition

نمایش اعداد

like 1's comp except shifted one position clockwise



È2 51m1

+

0 100 = + 4

1 100 = - 4

-

Only one representation for 0

One more negative number than positive number

نمایش اعداد

$N^* = 2^n - N$

$2^4 = 10000$

Example: Twos complement of 7

sub 7 = 0111

1001 = repr. of -7

Example: Twos complement of -7

$2^4 = 10000$

sub -7 = 1001

0111 = repr. of 7

Shortcut method:

Twos complement = bitwise complement + 1

0111 -> 1000 + 1 -> 1001 (representation of -7)

1001 -> 0110 + 1 -> 0111 (representation of 7)

مكمل 2

- **Here's an easier way to compute the *2's* complement:**
  1. Leave all least significant 0's and first 1 unchanged.
  2. Replace 0 with 1 and 1 with 0 in all remaining higher significant bits.

- Examples:

complement | unchanged

- N = 1010
  01 | 10

2's complement

complement | unchanged

N = 01011000
10101000

2's complement

جمع و تفريق
مكمل 2

4	0100	-4	1100
+ 3	0011	+ (-3)	1101
7	0111	-7	11001

If
(carry-in to sign = carry-out)
then ignore carry

if
(carry-in ≠ carry-out)
then overflow

4	0100	-4	1100
- 3	1101	+ 3	0011
1	10001	-1	1111

Simpler addition scheme makes twos complement the most common choice for integer number systems

جمع و تفريق
مكمل 2

Why can the carry-out be ignored?

-M + N when N > M:

$$M^* + N = (2^n - M) + N = 2^n + (N - M)$$

Ignoring carry-out is just like subtracting 2^n

-M + -N where N + M < or = 2^{n-1}

$$-M + (-N) = M^* + N^* = (2^n - M) + (2^n - N)$$

$$= 2^n - (M + N) + 2^n$$

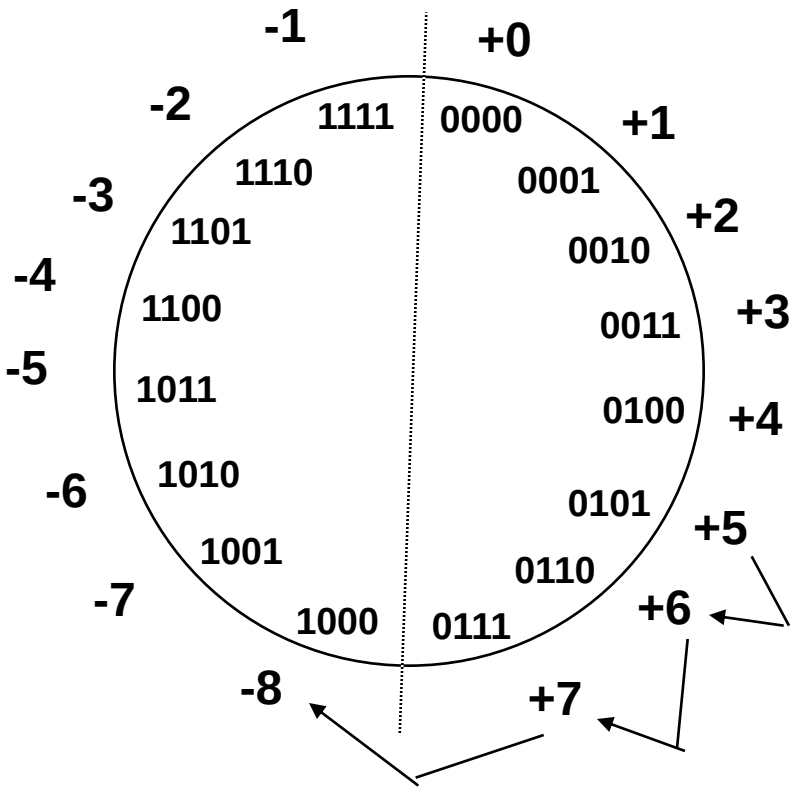
After ignoring the carry, this is just the right twos compl. representation for -(M + N)
or (M+N)*

سرریز

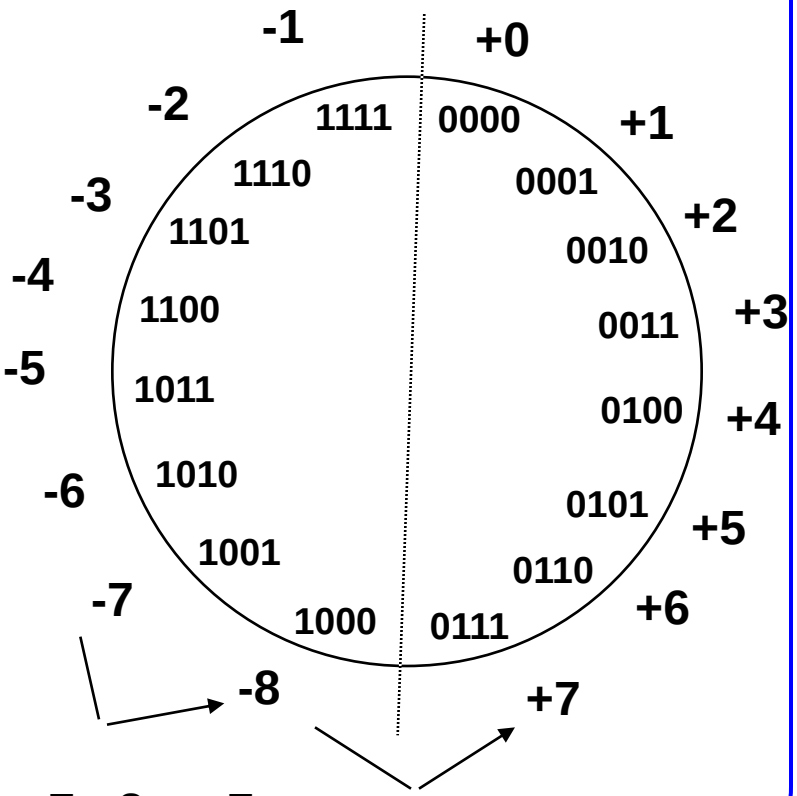
Overflow Conditions

Add two positive numbers to get a negative number

or two negative numbers to get a positive number



5 + 3 = -8



-7 - 2 = +7

سرریز

Overflow Conditions

	0 1 1 1
5	0 1 0 1
3	0 0 1 1
-8	1 0 0 0

Overflow

	1 0 0 0
-7	1 0 0 1
-2	1 1 1 0
7	1 0 1 1 1

Overflow

	0 0 0 0
5	0 1 0 1
2	0 0 1 0
7	0 1 1 1

No overflow

	1 1 1 1
-3	1 1 0 1
-5	1 0 1 1
-8	1 1 0 0 0

No overflow

Method 1: **Overflow when carry in to sign ≠ carry out**

Method 2: **Overflow when sign(A) = sign(B) ≠ sign (result)**

The Binary Multiplication

```
              1 0 1 0 1 0
×               1 0 1 1
-----------------------------
            1 0 1 0 1 0
          1 0 1 0 1 0
        0 0 0 0 0 0
+     1 0 1 0 1 0
-----------------------------
      1 1 1 0 0 1 1 1 0
```

Partial Products

ضرب باينري

- **Shift-and-add algorithm, as in base 10**

M’cand	0	0	0	1	1	0	1
M’plier	0	0	0	0	1	1	0
(1)			0	0	0	0	0
(2)		0	1	1	0	1	
(3)	0	1	1	0	1		
Sum	1	0	0	1	1	1	0

- **Check: 13 * 6 = 78**

Binary-Coded Decimal (BCD)

- A decimal code: Decimal numbers (0..9) are coded using 4-bit distinct binary words

- Observe that the codes 1010 .. 1111 (decimal 10..15) are NOT represented (invalid BCD codes)

TABLE 1-3
Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Table 1-3 Binary-Coded Decimal (BCD)

Binary-Coded Decimal

- **To code a number with *n* decimal digits, we need *4n* bits in BCD**

  e.g. $(365)_{10} = (0011\ 0110\ 0101)_{BCD}$

- **This is different from converting to binary, which is $(365)_{10} = (101101101)_2$**

- **Clearly, BCD requires more bits. BUT, it is easier to understand/interpret**

Application

BCD Addition

Case 1:

0001	1
0101	5
(0) 0110	(0) 6

Case 2:

0110	6
0101	5
(0) 1011	(1) 1

WRONG!

Case 3:

1000	8
1001	9
(1) 0001	(1) 7

Note that for cases 2 and 3, adding a factor of 6 (0110) gives us the correct result.

BCD Addition (cont.)

- **BCD addition is therefore performed as follows**
  1) Add the two BCD digits together using normal binary addition
  2) Check if correction is needed
     a) 4-bit sum is in range of 1010 to 1111
     b) carry out of MSB = 1
  3) If correction is required, add 0110 to 4-bit sum to get the correct result;

     → **BCD carry out = 1**

BCD Addition (cont.)

- **Example: Add 448 and 489 in BCD.**

```
 0100 0100  1000   (448 in BCD)
 0100 1000  1001   (489 in BCD)
            10001  (greater than 9, add 6)
            0110
            10111  (carry 1 into middle digit)
      1101         (greater than 9, add 6)
      0110
     10011         (carry 1 into leftmost digit)
1001 0011  0111    (BCD coding of 937₁₀)
```

BCD Negative Number Representation

- **Similar to binary negative number representation but for r = 10.**
  - BCD 9's complement
    - invert each BCD digit ($0 \rightarrow 9$, $1 \rightarrow 8$, $2 \rightarrow 7$, $3 \rightarrow 6$, …$7 \rightarrow 2$, $8 \rightarrow 1$, $9 \rightarrow 0$)
  - BCD 10's complement
    - $-N \equiv 10^n - N$; 9's complement + 1

Excess-3

- مانند BCD ولي هر رقم +3
  - جمع سرراست تر
  - self-comlpement code
    - (مکمل هر رقم = مکمل 9 آن)

Decimal Digit	BCD				Excess-3			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

ASCII character code

- We also need to represent letters and other symbols → alphanumeric codes
- ASCII = American Standard Code for Information Interchange. Also known as Western European
- It contains 128 characters:
  - 94 printable (26 upper case and 26 lower case letters, 10 digits, 32 special symbols)
  - 34 non-printable (for control functions)
- Uses 7-bit binary codes to represent each of the 128 characters

ASCII Table

		$b_6b_5b_4$ (column)							
$b_3b_2b_1b_0$	Row (hex)	000 0	001 1	010 2	011 3	100 4	101 5	110 6	111 7
0000	0	NUL Null	DLE	SP Space	0	@	P	`	p
0001	1	SOH	DC1	!	1	A	Q	a	q
0010	2	STX	DC2	"	2	B	R	b	r
0011	3	ETX	DC3	#	3	C	S	c	s
0100	4	EOT	DC4	$	4	D	T	d	t
0101	5	ENQ	NAK	%	5	E	U	e	u
0110	6	ACK	SYN	&	6	F	V	f	v
0111	7	BEL Bell	ETB	'	7	G	W	g	w
1000	8	BS BkSpc	CAN	(	8	H	X	h	x
1001	9	HT Tab	EM	)	9	I	Y	i	y
1010	A	LF Line Fd	SUB	*	:	J	Z	j	z
1011	B	VT	ESC Escape	+	;	K	[	k	{
1100	C	FF	FS	,	<	L	\	l	|
1101	D	CR Crg Ret	GS	-	=	M	]	m	}
1110	E	SO	RS	.	>	N	^	n	~
1111	F	SI	US	/	?	O	_	o	DEL

ASCII Control Codes

Control codes

NUL	Null	DLE	Data link escape
SOH	Start of heading	DC1	Device control 1
STX	Start of text	DC2	Device control 2
ETX	End of text	DC3	Device control 3
EOT	End of transmission	DC4	Device control 4
ENQ	Enquiry	NAK	Negative acknowledge
ACK	Acknowledge	SYN	Synchronize
BEL	Bell	ETB	End transmitted block
BS	Backspace	CAN	Cancel
HT	Horizontal tab	EM	End of medium
LF	Line feed	SUB	Substitute
VT	Vertical tab	ESC	Escape
FF	Form feed	FS	File separator
CR	Carriage return	GS	Group separator
SO	Shift out	RS	Record separator
SI	Shift in	US	Unit separator
SP	Space	DEL	Delete or rubout

Unicode

- Established standard (16-bit alphanumeric code) for international character sets
- Since it is 16-bit, it has 65,536 codes
- Represented by 4 Hex digits
- ASCII is between 0000_{16} .. $007B_{16}$

Unicode Table

http://www.unicode.org/charts/

Unicode

062B 1579	ث 062C 1580	ج 062D 1581	ح 062E 1582	خ
0633 1587	س 0634 1588	ش 0635 1589	ص0636 1590	ض
063B 1595	ڭ 063C 1596	ڭ 063D 1597	ئ 063E 1598	ئ
0643 1603	ك 0644 1604	ل 0645 1605	م 0646 1606	ن
064B 1611	ً 064C 1612	ٌ 064D 1613	ٍ 064E 1614	َ
0653 1619	ٓ 0654 1620	ٔ 0655 1621	ٕ 0656 1622	ِ
065B 1627	ٛ 065C 1628	ٜ 065D 1629	ٞ 065E 1630	ّ
0663 1635	٣ 0664 1636	٤ 0665 1637	٥ 0666 1638	٦
066B 1643	, 066C 1644	, 066D 1645	* 066E 1646	ل
0673 1651	إ 0674 1652	٭ 0675 1653	ٱ 0676 1654	ؤ
067B 1659	ٻ 067C 1660	ٽ 067D 1661	ٿ 067E 1662	پ
0683 1667	ڃ 0684 1668	ڄ 0685 1669	څ 0686 1670	چ
068B 1675	ڊ 068C 1676	ڌ 068D 1677	ڍ 068E 1678	ڏ

ASCII Parity Bit

- Parity coding is used to detect errors in data communication and processing
  - An 8th bit is added to the 7-bit ASCII code
- <u>Even (Odd)</u> parity: set the parity bit so as to make the # of 1's in the 8-bit code even (odd)

ASCII Parity Bit (cont.)

- **For example:**
  - Make the 7-bit code 1011011 an 8-bit even parity code → 11011011
  - Make the 7-bit code 1011011 an 8-bit odd parity code → 01011011
- **Error Checking:**
  - Both even and odd parity codes can detect an odd number of error.
    - An even number of errors goes undetected.

Gray Codes

- **Gray codes are *minimum change* codes**
  - From one numeric representation to the next, only one bit changes
  - Applications:
    - Later.

Gray Codes (cont.)

Binary	Gray
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

Binary	Gray
00	00
01	01
10	11
11	10

Binary	Gray
000	000
001	001
010	011
011	010
100	110
101	111
110	101
111	100